

CS-499: 5-2 Milestone Four: Enhancement Three: Databases

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Briefly describe the artifact. What is it? When was it created?

The artifact I have once again chosen for this enhancement is the Travlr Getaways Full-Stack Web Application, which was created in my previous term, CS 465: Full Stack Development 1. The project, put simply, is a full-stack travel booking application built using the MEAN stack, consisting of an Angular administrator application, a Node.js and Express backend, and a MongoDB database accessed through Mongoose. The application also uses JWT authentication, which enables admin access management, as well as follows the MVC architecture, which separates the routes and controllers.

A month back, when submitting the application as a final project, it was considered finished, and with the perfect grade coming back, it was considered exceptional. Going through this capstone with the opportunity to revisit the project from a professional software developer's perspective, I can see how there was still much to be done, improvements which in hindsight look as though they should have been there from the initial building of the application. With the previous enhancements focusing on software engineering and algorithmic improvements, this final one was focusing more on what I could do to strengthen the applications database design. This was done by improving schema validation, data integrity, indexing, and database query performance, which makes the application's data more reliable, but also improves the efficiency of storing and retrieving information.

Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

I decided to choose this artifact once again; this was done because it shows much more than database concepts by themselves. This same artifact throughout this capstone has allowed me to showcase not just software engineering, but also algorithms and database development, all within a single application, rather than presenting several unrelated assignments. I believe showcasing one final application in the end, which slowly improved over multiple milestones, better represents how software is actually developed and maintained in the industry.

For this enhancement, my goal was to improve the applications database layer, instead of adding visible user features, and to do that, I strengthened the Trip schema by improving its indexing strategy, enforcing unique trip codes, enforces unique trip codes, trimmed unneeded white space, made better validation rules, stored trip codes in uppercase automatically, and finally added minimum and maximum length validation where it was appropriate to do so. Doing these improvements helped maintain consistent data, which also reduces the possibility of invalid records entering the database.

I also went ahead and improved the way the application can communicate with MongoDB, where Read-only queries now use Mongoose's `lean()` method, which reduces unnecessary overhead, and unnecessary database fields are excluded from the API responses using `select()`. With the work done in previous milestones like searching, sorting, and pagination, the database enhancements are in place to improve query performance while making the application more scalable as more and more trip records are added.

Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

I believe this enhancement served as a proper finish and met the remaining course outcomes that I planned to address during Module One, with the primary outcome shown in this enhancement being the ability to demonstrate well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals. In this enhancement, I strengthened the applications database by improving schema validation, indexing, data integrity, and query performance, and in doing so, I increased the reliability and maintainability of the application while showing professional database practices.

This enhancement continued to strengthen my understanding of how it is to develop secure software by making strict validation rules within the schema, preventing duplicate records, and improving the consistency of stored data, where the database itself now helps to protect the integrity of the application, instead of relying only on the application logic.

With this enhancement finally done, I do believe that I have successfully addressed the outcome coverage plan which was talked about in Module One. Across the three milestones and enhancements done on the same artifact, I was able to improve the application's software engineering practices, implement algorithmic enhancements such as searching, sorting, and pagination, and strengthen the database, and all together these enhancements show the complete progression from what was once a simple final class project to what is now a more professional full-stack application.

Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

When I initially began planning this enhancement, I assumed it would simply be creating collections and storing records, as when I originally created this project for CS465, the objective was making sure information was able to be saved and retrieved successfully, but during this capstone I realized that designing a database is much more than that, and the structure of a database has a much bigger impact on performance, maintainability and the quality of the overall application.

I didn't realize how much responsibility belongs within the database itself, where originally much of the validation depended on the application, but by strengthening the Mongoose schema with indexes, unique constraints, and additional validation rules, I was able to improve the overall integrity of the stored data before it was ever going to reach the database. I also better understood how query optimizations, such as using `lean()` for read-only operations, can improve efficiency without changing how the application behaves for the user from their perspective.

That isn't to say that this enhancement didn't come with challenges of its own, with this being the final enhancement, I had to attempt to make the database improvements without breaking the functionality that had already been completed in the prior milestones, and since the engineering and algorithm enhancements depended on the same database structure, after every minor modification I carefully tested again and again to make sure that searching, sorting, pagination, authentication, and CRUD operations were still functioning as they were supposed to. When you're so close to the end, you cannot risk doing something so major that you risk breaking everything you worked on previously, keeping in mind how even relatively small database changes can affect the rest of the system. Looking back at all the enhancements I chose to do, I feel as if this capstone has transformed my original project into a much more complete representation of what I am capable of, and better shows the development skills I have learned throughout this Computer Science program.